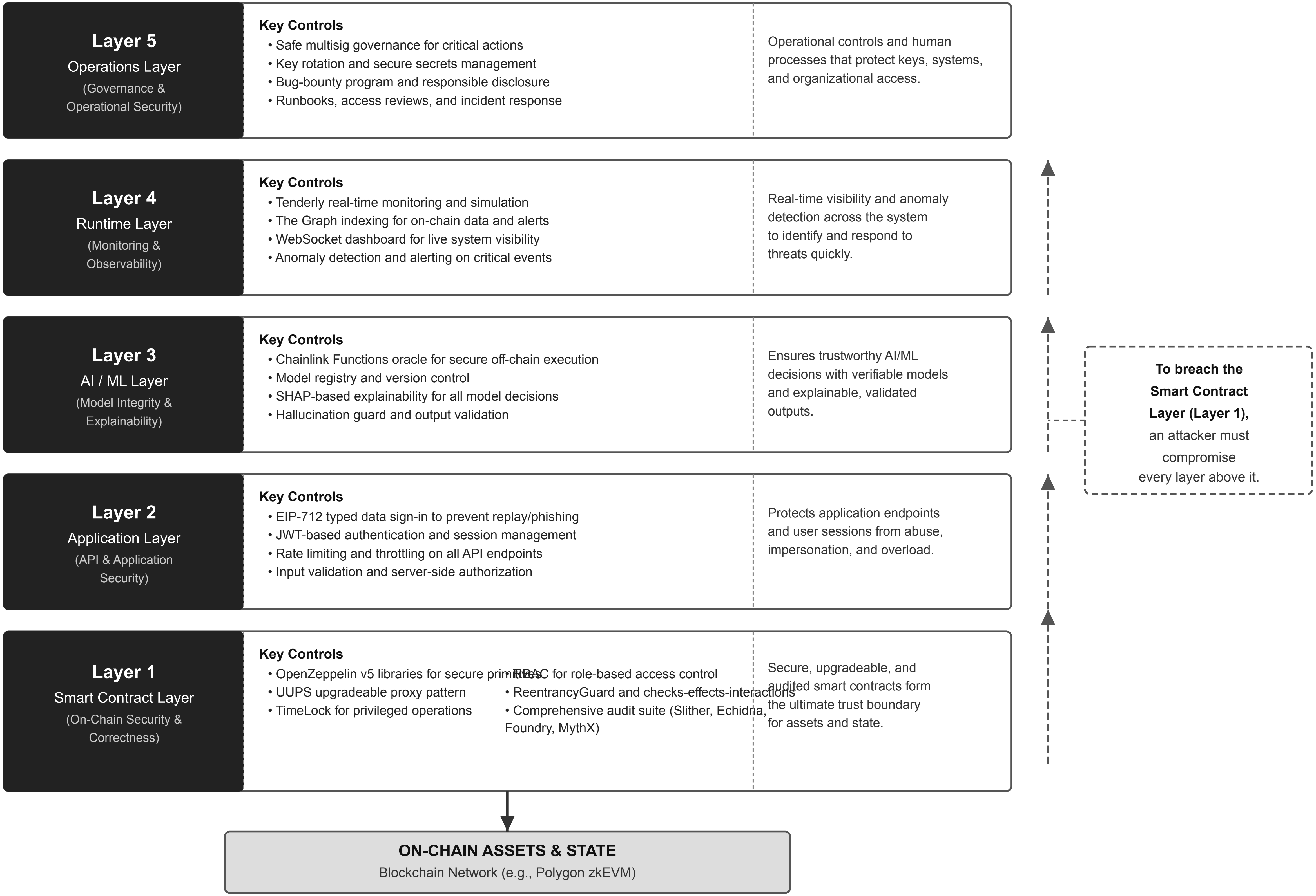
Layer 5
Operations Layer
(Governance & Operational Security)
Key Controls
• Safe multisig governance for critical actions
• Key rotation and secure secrets management
• Bug-bounty program and responsible disclosure
• Runbooks, access reviews, and incident response
Operational controls and human processes that protect keys, systems, and organizational access.
Layer 4
Runtime Layer
(Monitoring & Observability)
Key Controls
• Tenderly real-time monitoring and simulation
• The Graph indexing for on-chain data and alerts
• WebSocket dashboard for live system visibility
• Anomaly detection and alerting on critical events
Real-time visibility and anomaly detection across the system to identify and respond to threats quickly.
Layer 3
AI / ML Layer
(Model Integrity & Explainability)
Key Controls
• Chainlink Functions oracle for secure off-chain execution
• Model registry and version control
• SHAP-based explainability for all model decisions
• Hallucination guard and output validation
Ensures trustworthy AI/ML decisions with verifiable models and explainable, validated outputs.
Layer 2
Application Layer
(API & Application Security)
Key Controls
• EIP-712 typed data sign-in to prevent replay/phishing
• JWT-based authentication and session management
• Rate limiting and throttling on all API endpoints
• Input validation and server-side authorization
Protects application endpoints and user sessions from abuse, impersonation, and overload.
Layer 1
Smart Contract Layer
(On-Chain Security & Correctness)
Key Controls
• OpenZeppelin v5 libraries for secure primitives
• UUPS upgradeable proxy pattern
• TimeLock for privileged operations
• RBAC for role-based access control
• ReentrancyGuard and checks-effects-interactions
• Comprehensive audit suite (Slither, Echidna, Foundry, MythX)
Secure, upgradeable, and audited smart contracts form the ultimate trust boundary for assets and state.
To breach the Smart Contract Layer (Layer 1), an attacker must compromise every layer above it.
ON-CHAIN ASSETS & STATE
Blockchain Network (e.g., Polygon zkEVM)